

Item(item_id, title, creator, publisher, year, subject_genre, language) FD: item_id → title, creator, publisher, year, subject_genre, language item_id* = all attributes → superkey **In BCNF**

PrintBook(item_id, ISBN, page_num) item_id → item FD: item_id → ISBN, page_num item_id* = {item_id, ISBN, page_num} = all attributes → superkey FD: ISBN → item_id, page_num ISBN* = {ISBN, item_id, page_num} = all attributes → superkey **In BCNF**

EBook(item_id, ISBN, format) item_id → item FD: item_id → ISBN, format item_id* = {item_id, ISBN, format} = all attributes → superkey FD: ISBN → item_id, format ISBN* = {ISBN, item_id, format} = all attributes → superkey **In BCNF**

Magazine(item_id, ISSN, issue, issue_date) item_id → item FD: item_id → ISSN, issue, issue_date item_id* = all attributes → superkey FD: (ISSN, issue) → item_id, issue_date (ISSN, issue)* = all attributes → superkey ISSN alone is not a determinant **In BCNF**

Journal(item_id, ISSN, issue, volume) item_id → item FD: item_id → ISSN, issue, volume item_id* = all attributes → superkey FD: (ISSN, volume, issue) → item_id (ISSN, volume, issue)* = all attributes → superkey ISSN alone is not a determinant **In BCNF**

Record(item_id, type, quantity) item_id → item FD: item_id → type, quantity item_id* = all attributes → superkey **In BCNF**

Copy(item_id, copy_number, status) item_id → item FD: (item_id, copy_number) → status (item_id, copy_number)* = all attributes → superkey **In BCNF**

Loan(loan_id, checkout_date, due_date, returned_date, item_id, copy_number, member_id) (item_id, copy_number) → Copy, member_id → Member FD: loan_id → checkout_date, due_date, returned_date, item_id, copy_number, member_id loan_id* = all attributes → superkey FD: (item_id, copy_number, checkout_date) → loan_id (item_id, copy_number, checkout_date)* = all attributes → superkey **In BCNF**

Fine(loan_id, date_issued, amount, status) loan_id → Loan FD: loan_id → date_issued, amount, status loan_id* = all attributes → superkey **In BCNF**

Member(member_id, name, address, phone, email, reg_date, status) FD: member_id → name, address, phone, email, reg_date, status member_id* = all attributes → superkey FD: email → member_id, name, address, phone, reg_date, status email* = all attributes → superkey **In BCNF**

Hold(member_id, item_id, date_placed) member_id → Member, item_id → Item FD: (member_id, item_id) → date_placed (member_id, item_id)* = all attributes → superkey **In BCNF**

Donates(item_id, copy_number, member_id, date_donated) (item_id, copy_number) → Copy, member_id → Member FD: (item_id, copy_number) → member_id, date_donated (item_id, copy_number)* = all attributes → superkey **In BCNF**

Room(room_id, name, capacity) FD: room_id $\rightarrow$ name, capacity room_id⁺ = all attributes $\rightarrow$ superkey **In BCNF**

Event(event_id, name, date, room_id, type, time, audience, capacity) room_id $\rightarrow$ Room FD: event_id $\rightarrow$ name, date, room_id, type, time, audience, capacity event_id⁺ = all attributes $\rightarrow$ superkey FD: (room_id, date, time) $\rightarrow$ event_id (room_id, date, time)⁺ = all attributes $\rightarrow$ superkey **In BCNF**

Registers(member_id, event_id, reg_date) member_id $\rightarrow$ Member, event_id $\rightarrow$ Event FD: (member_id, event_id) $\rightarrow$ reg_date (member_id, event_id)⁺ = all attributes $\rightarrow$ superkey **In BCNF**

Volunteers(member_id, event_id) member_id $\rightarrow$ Member, event_id $\rightarrow$ Event All-key, no non-trivial FDs **In BCNF**

Personnel(employee_id, first_name, last_name, role, email, phone) FD: employee_id $\rightarrow$ first_name, last_name, role, email, phone employee_id⁺ = all attributes $\rightarrow$ superkey **In BCNF**

Manages(employee_id, item_id) employee_id $\rightarrow$ Personnel, item_id $\rightarrow$ Item All-key, no non-trivial FDs **In BCNF**